

Lecture 1

Overview of Natural Language Processing

What the field is, why language resists computation, and how this course is built

Intro duction to Natural Lang uage Process ing

How a modern tokenizer actually reads the title of this course

What we cover today

01

What NLP is

Definition, scope, how it sits against computational linguistics and text mining, and the split between understanding and generation.

02

Why language is hard

Ambiguity at every level, variability, context dependence, and a frequency distribution with no end to it.

03

How the field is organised

Six levels of linguistic analysis, six shapes that almost every task takes, and the shape of a working NLP system.

04

How this course is built

The fifteen-week map, what you need coming in, and the tools we will use.

What is Natural Language Processing?

The study of computational methods that let machines analyse, interpret, and generate human language.

Linguistics

- The structure of language
- Morphology, syntax, semantics
- Supplies the categories we model

Computer Science

- Algorithms and data structures
- Efficiency and scale
- Supplies the machinery

Machine Learning

- Learning patterns from data
- Statistical inference
- Supplies the modern engine

Three neighbouring labels, often confused

Computational Linguistics

Uses computation to test theories about how language works. Science first.

Natural Language Processing

Builds systems that do useful things with language. Engineering first.

Text Mining

Extracts patterns and facts from large text collections. Often a consumer of NLP.

Two directions: understanding and generation

Natural Language Understanding

Input is language. Output is structure, a label, or a decision.

- Is this review positive or negative?
- Which words in this sentence name a person?
- Does sentence A imply sentence B?
- What is the grammatical structure here?

Natural Language Generation

Output is language. Input may be data, meaning, or other text.

- Translate this paragraph into Urdu
- Summarise this report in five lines
- Answer this question in plain language
- Continue this conversation sensibly

A distinction that is useful, and increasingly blurred

A modern generative model performs classification by generating a label as text, so the architectural boundary has largely dissolved. The conceptual boundary still earns its keep, because the two directions fail differently and are evaluated differently. A classifier can be wrong; a generator can be wrong, fluent, and confident at the same time. We will treat them separately until Week 12, and then watch them merge.

Where NLP already runs your day

1

Machine Translation

Sentence and document translation across many language pairs.

2

Search & Retrieval

Query understanding, ranking, and semantic retrieval.

3

Conversational AI

Assistants, support bots, task-oriented dialogue.

4

Summarization

Condensing long documents into something readable.

5

Information Extraction

Entities, relations and events pulled into structured records.

6

Sentiment & Opinion

Polarity, stance, and aspect-level opinion at scale.

7

Clinical & Biomedical

Coding notes, cohort selection, literature triage.

8

Code & Formal Text

Program synthesis, legal review, financial analysis.

Ambiguity is the default, not the exception

Lexical

I went to the bank.

Financial institution, or river edge? One form, unrelated senses.

Part of speech

Time flies like an arrow.

Several valid parses. 'Flies' is a verb or a noun; 'like' a preposition or a verb.

Attachment

I saw the man with the telescope.

Did I use the telescope, or does he have one? Syntax alone cannot decide.

Quantifier scope

Every student read a book.

One book that all read, or one book each? Two different logical forms.

Coreference

The trophy did not fit in the suitcase because it was too big.

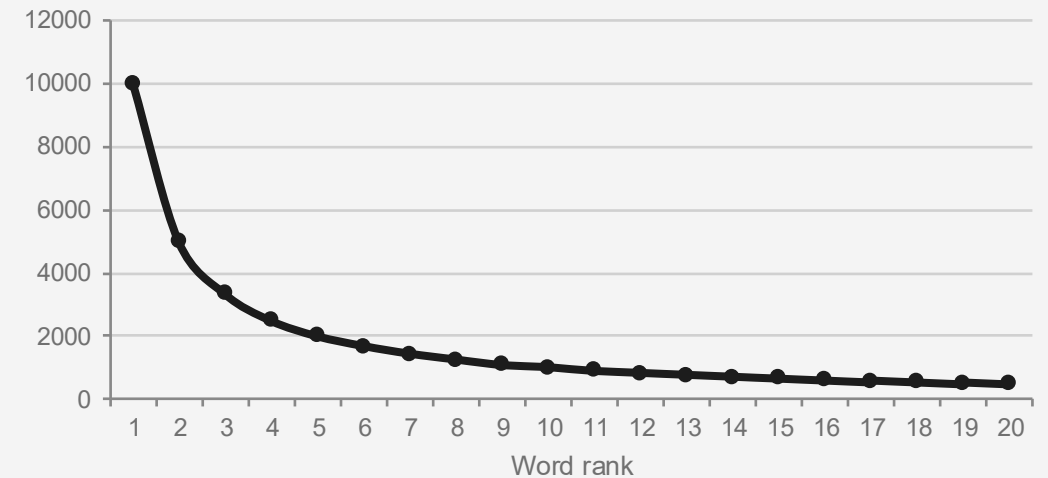
What does 'it' refer to? Swap 'big' for 'small' and the answer flips.

Variability, context, and a very long tail

Four further obstacles

- Variability — one meaning, unbounded forms. 'The match was postponed' and 'They called off the game' share no content words.
- Context dependence — 'here', 'yesterday', 'she' cannot be resolved from the sentence alone. Meaning is indexical.
- Pragmatics — 'Can you pass the salt?' is a request, not a question about ability. Literal semantics gives the wrong answer.
- Non-compositionality — 'kick the bucket' is not about buckets. Idiom, metaphor and sarcasm break the compositional assumption.

Productivity: Zipf's law



Word frequency falls off as roughly $1/\text{rank}$. A handful of function words dominate every corpus, while most of the vocabulary appears once or twice. Whatever corpus you train on, the next document will contain words you have never seen. This is why open-vocabulary handling is not optional, and it is the reason Week 3 exists.

One sentence, and everything a machine must decide

“Dr. Khan could not treat the patient because she was unavailable.”

1 Where does the sentence end?

The full stop after 'Dr' is an abbreviation, not a boundary. Naive splitting produces two broken sentences.

2 What are the units?

'could not' is two tokens; a contraction would be one string and two units. This decision changes every downstream count.

3 What is the structure?

'because' introduces a subordinate clause. The parser must attach it to the right verb.

4 Which sense of 'treat'?

Medical care, not hospitality and not processing a material. Context selects among unrelated senses.

5 Who is 'she'?

Dr. Khan or the patient. Both are grammatically available. Only world knowledge about why treatment fails resolves it.

6 What is actually meant?

A scheduling failure is being reported, and possibly an excuse offered. The literal content understates the message.

Six levels of linguistic analysis

1	Phonology / Orthography	Sound and writing systems	<i>Script normalization, transliteration</i>	Week 3
2	Morphology	Internal structure of words	<i>Stemming, lemmatization, segmentation</i>	Week 3
3	Syntax	How words combine into phrases	<i>POS tagging, dependency parsing</i>	Week 6
4	Semantics	Literal meaning of expressions	<i>Word-sense disambiguation, similarity</i>	Weeks 5-6
5	Pragmatics	Meaning in the context of use	<i>Intent detection, implicature</i>	Week 9
6	Discourse	Structure across sentences	<i>Coreference, coherence, summarization</i>	Weeks 6, 12

Tone groups the levels by kind: form (light), structure and meaning (mid), use and context (dark). The right-hand column is where each is covered in this course.

Almost every NLP task is one of six shapes

1 Text classification

Sequence → one label

Sentiment, topic, spam, intent

Week 8

2 Sequence labelling

Sequence → label per token

POS tagging, NER, chunking

Week 6

3 Span extraction

Sequence → (start, end)

Extractive question answering

Week 12

4 Sequence to sequence

Sequence → sequence

Translation, summarization, paraphrase

Weeks 10-13

5 Structured prediction

Sequence → tree or graph

Dependency parsing, relation extraction

Week 6

6 Open-ended generation

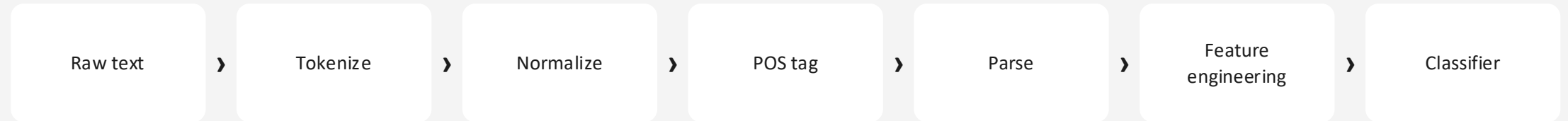
Prompt → free text

Dialogue, code, long-form answers

Weeks 12-14

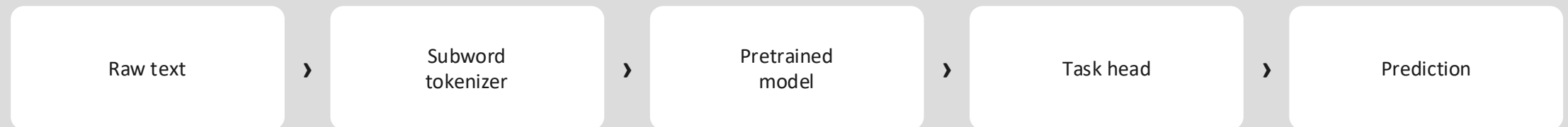
What an NLP system actually looks like

Classical pipeline · explicit, modular, error-propagating



Each stage is built separately. A tagging error at stage four is invisible to stage seven, which simply receives a wrong feature. Weeks 3 to 8 build this pipeline, stage by stage.

Modern pipeline · end-to-end, learned, jointly optimised



Morphology, syntax and semantics are still computed, but as internal representations learned from data rather than named, inspectable stages. Weeks 10 to 14 build this one.

Weeks 1 to 8: foundations and classical NLP

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|---|-----------|---|
| 1 | Lec 1-2 | Overview of NLP; History of the field |
| 2 | Lec 3-4 | Setting up the NLP environment; Python crash course and core libraries |
| 3 | Lec 5-6 | Tokenization, words and tokens, regular expressions; cleaning, stop words, normalization, stemming, lemmatization |
| 4 | Lec 7-8 | Word representations: one-hot, bag of words, TF-IDF, n-grams, Word2Vec, CBOW, skip-gram; pretrained embeddings: GloVe, fastText, BERT |
| 5 | Lec 9-10 | Text similarity: edit-, token- and sequence-based algorithms; cosine and soft cosine similarity, applications |
| 6 | Lec 11-12 | Part-of-speech tagging, named entity recognition, dependency parsing; word sense disambiguation |
| 7 | Lec 13-14 | Classical language models: n-grams, perplexity, smoothing |
| 8 | Lec 15-16 | Text classification: naive Bayes, logistic regression, SVM; design decisions and building a dataset |

Mid-term examination week follows Week 8

Weeks 9 to 15: neural NLP and the modern stack

9

Lec 17-18

Sentiment analysis and evaluation metrics; paper presentation session

10

Lec 19-20

Introduction to deep learning for NLP; sequence models

11

Lec 21-22

Sequence models in depth: RNN, LSTM, GRU

12

Lec 23-24

Language models and a hands-on build; transformers, attention, BERT, GPT

13

Lec 25-26

Pre-training and fine-tuning with GPT; prompt engineering and parameter-efficient adaptation

14

Lec 27-28

Advanced NLP: retrieval-augmented generation and agents; multimodal language models

15

Lec 29-30

Ethical challenges and real-world implications; responsible AI

Week 9's paper session is yours: you present a recent NLP paper to the class. Start looking for one now, not in Week 8.

What you need coming in

Python

Essential

Functions, classes, list and dict comprehensions, reading files, installing packages. If this is shaky, Week 2 is your safety net, but it is a crash course, not a first course.

Probability

Essential

Conditional probability, Bayes' rule, independence, expectation. Almost every model in Weeks 7 to 9 is a probability estimate wearing different clothes.

Linear algebra

Essential

Vectors, matrices, dot products, cosine of the angle between two vectors. Week 4 turns words into vectors and never looks back.

Machine learning

Strongly preferred

Train/validation/test splits, overfitting, gradient descent, precision and recall. We will revise these, but we will not teach them from zero.

Calculus

Helpful

Partial derivatives and the chain rule, enough to follow what backpropagation does. Needed only from Week 10 onward.

Command line

Helpful

Navigating directories, running scripts, managing virtual environments, basic git. Week 2 covers the setup, but comfort here saves you hours.

The toolkit for this semester

Environment

- Python 3.11 or later
- conda or venv for isolation
- Jupyter for exploration
- git and a GitHub account

Classical NLP

- NLTK for teaching and classical algorithms
- spaCy for fast production pipelines
- scikit-learn for baselines and metrics
- Gensim for Word2Vec and topic models

Neural NLP

- PyTorch as the deep learning framework
- Hugging Face Transformers, Datasets, Tokenizers
- Kaggle Notebooks or Colab for free GPU time

Two things to sort out before Thursday

Install Python and confirm it runs from your terminal, and create a Kaggle account and verify your phone number so the free GPU quota is available. Nobody needs to buy hardware for this course: every assignment is sized to run on a free notebook service, and the few that benefit from a GPU fit comfortably inside the free weekly allowance.

Five things to take away

1

NLP is the engineering of language understanding and generation

It borrows categories from linguistics, machinery from computer science, and its current engine from machine learning.

2

Language is hard for structural reasons

Ambiguity at every level, unbounded variability, dependence on context, and a frequency distribution with no end. These are properties of language, not failures of our methods.

3

The field organises the problem in two ways

Six levels of linguistic analysis, and six shapes that nearly every task takes. Learn to identify both on sight.

4

There are two ways to build a system

An explicit pipeline you can inspect, and an end-to-end model you mostly cannot. This course builds the first, then the second.

5

Ask what a number means before you trust it

This will be the recurring theme of the course. A metric that looks rigorous can still be measuring almost nothing.

Lecture 2: History of the field

What we will cover

- The symbolic era: hand-written grammars, ELIZA, SHRDLU, and why rule-based systems stalled
- The statistical turn: corpora, annotation, hidden Markov models, and the IBM translation work
- The neural shift: learned representations replacing engineered features
- The pretraining era: one backbone, many tasks
- What each transition actually changed, and what it cost

To do before Thursday

- Read Jurafsky & Martin, Speech and Language Processing, 3rd edition, Chapter 1. It is free online.
- Install Python 3.11+ and confirm it runs from your terminal
- Create and verify a Kaggle account
- Write down one NLP application you use regularly and one thing about it that annoys you. We will use these in Week 8.

Primary textbook for the course

Jurafsky & Martin, Speech and Language Processing (3rd edition draft), free online and updated regularly. Supplementary: Eisenstein, Introduction to Natural Language Processing; Goldberg, Neural Network Methods for NLP. Lecture videos: Stanford CS224N. Every paper we discuss is free through the ACL Anthology.